

Little Red Dots: Super-Eddington Stellar Envelopes

Type IIIn supernovae and eta Carinae's Great Eruption offer a unified model for JWST's Little Red Dots.

arXiv:2606.30711 — Little Red Dots as intermediate mass, super-Eddington engines.

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Background

Little Red Dots (LRDs) show a unique mix of features not seen together in any other galaxy or AGN. Their nature remains debated, with some claiming overmassive black holes.

No self-consistent explanation exists for LRDs' blackbody-like continua, broad lines, and lack of variability.

What this paper is selling

The authors draw parallels to Type IIIn supernovae and eta Carinae's Great Eruption, where dense winds create pseudo-photospheres. They propose LRDs are intermediate-mass, super-Eddington engines enshrouded in slow winds.

Key results to remember

- LRDs, Type IIIn SNe, and eta Carinae share similar temperatures and line profiles.
- Engine mass < 10⁵ solar masses, likely 10³-6, super-Eddington.
- Dust production may seed classical AGN, linking life cycles.

